

Reproducing and Extending TISER: Context–Memory Conflict and Tennis-Domain Adaptation for Temporal Reasoning

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Abstract—We reproduce TISER, which elicits reasoning, a timeline, reflection, and an answer, and study two extensions. A context–memory benchmark tests responses to evidence that contradicts a TISER model’s stable recall; a synthetic tennis dataset tests continued domain adaptation. Aggregate in-domain reproduction is similar to the paper, despite a large TGQA EM discrepancy. TISER weights and prompting increase adherence to edited context, but reflections rarely name the conflict and order reversal remains difficult. On a reserved 113-item tennis holdout, continued adaptation improves EM from 0.575 to 0.735 over the original adapter, although the missing unadapted 7B baseline prevents attributing direct-transfer performance to TISER training. Retention is inconclusive.

I. Introduction

Temporal reasoning requires ordering, comparing, and measuring events. TISER [1] trains models to emit tagged reasoning, timeline, reflection, and answer stages. We reproduce its in-domain evaluation and study:

- 1) Context–memory conflict: an inference-only probe of adherence to edited evidence and explicit conflict acknowledgement.
- 2) Tennis adaptation: supervised LoRA transfer to synthetic tennis temporal QA.

We contribute an implementation-level Qwen2.5-7B reproduction, a controlled conflict benchmark, and a two-scale adaptation and retention study. Code and run records are in the repository.

II. Background and Problem Definition

A. TISER Formulation

For question q , context c , and answer a , TISER generates $y = (r, t, s, a)$: reasoning, timeline, reflection, and final answer. Only the extracted answer is scored. We evaluate TGQA [2], TempReason L2/L3 [3], and TimeQA easy/hard [4]; ToT-semantic is treated as OOD.

B. Research Questions

We ask whether the reproduction recovers the reported performance regime; whether model/prompt choice, conflict type, and reflection affect adherence to conflicting context; and whether continued tennis adaptation improves over the original adapter without erasing in-domain

performance. No unadapted 7B tennis run was available to isolate prior TISER training.

III. Methodology

A. Baseline Reproduction

We train Qwen2.5-Instruct [6] in the released TISER format and score the extracted answer with exact match (EM) and token F1.

B. Extension 1: Context–Memory Conflict

a) Memory elicitation and eligibility.: We remove context from real-entity items and retain those for which the TISER-trained model recalls the gold answer in at least three of five samples. The benchmark is therefore TISER-selected. The vanilla model independently recalls the same answer on only 206/537 conflict sources (38.4%); its other rows test contextual use, not conflict with its demonstrated memory.

b) Conflict construction.: Deterministic edits shift dates (C1), substitute an entity from the same broad question-type pool (C2), or reverse intervals (C3), while controls only reorder context. C2 does not guarantee fine-grained type compatibility. Every conflict changes the contextual answer relative to TISER recall.

c) Run matrix and measures.: Without retraining, a 2×2 matrix crosses vanilla/TISER weights with standard-/TISER prompts. Faithful EM scores the edited answer; original EM scores the pre-edit TISER recall. An audited faithful response that does not explicitly acknowledge disagreement is a silent override.

C. Extension 2: Tennis-Domain Adaptation

a) Dataset creation and processing.: GPT-5.5 generated synthetic tennis question–context–answer records and, separately, TISER training traces using the prompts in Appendix A. Duplicate groups remain within splits; only training records receive traces. Model audits check semantic answerability and trace support. Data are CC BY 4.0 and code is MIT licensed.

b) Adaptation strategies.: At 0.5B we compare prompting and tennis-only training; at 7B we compare the original TISER adapter with continued tennis adaptation and test original-domain retention.

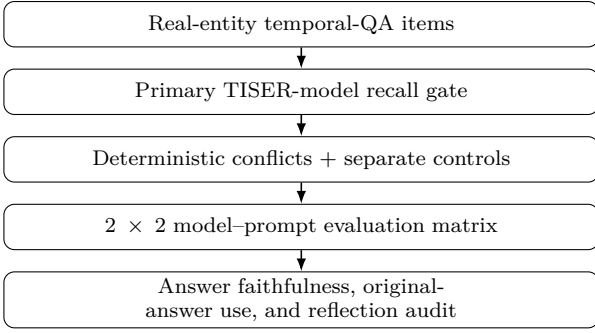


Fig. 1. Inference-only context-memory evaluation pipeline.

IV. Experimental Setup

A. Data and Evaluation

After a 2,048-token filter, TISER training uses 54,488 items; evaluation uses 19,214 in-domain and 2,800 OOD items. From 15,898 real-entity candidates, the conflict pipeline yields 1,056 conflicts and 120 controls.

B. Tennis Data, Trace, and Parsing Details

A seed-42, category- and duplicate-group-aware split assigns 785/224/113 of 1,122 records to training/selection/-final holdout. Of 785 training candidates, 50 pilot traces are excluded from reported training, 600 disjoint traces train both adapters, and 135 lack generated outputs. File repair wrapped 200 untagged traces and recovered 26 records from 19 malformed lines; semantic validity is audited separately. Evaluation extracts `<answer>` first, then applies documented fallbacks and tennis-specific normalization. Neither evaluation split has training traces or affects gradients; 224 items select among models, while 113 remain the final holdout, although complete historical non-use cannot be externally verified.

C. Tennis Semantic and Trace Audits

Two blinded, file-batch model audits cover 1,121 unique semantic payloads (all 1,122 rows) and all 650 traces. Two separate GPT-5.6 Sol/high-reasoning Codex tasks [8] label support, ambiguity/error, or unscorability using item-specific rationales and exact evidence; a third adjudicates 9 semantic and 12 trace disagreements plus proposed gold changes. Hash and schema checks establish provenance, not correctness. The audits do not alter training data, use one uncalibrated model family, and lack verifiable backend/decoding metadata.

D. Models, Training, and Decoding

Models generate all tags autoregressively; prompt tokens are masked from loss. LoRA [5] targets attention $q/k/v/o$. Reproduction is unquantized; continued 7B adaptation uses 4-bit weights. Greedy decoding permits 768 new tokens, except 256 for 7B tennis evaluation.

The selected 7B adapter is the best of 22 candidates on the common 224-item split (1–3 epochs, three learning

TABLE I
Training settings; Micro. $\times$ GA is microbatch by accumulation.

Run	n	Ep.	Updates	Micro. $\times$ GA	Quant.
TISER 7B	54,488	3	10,215	8 $\times$ 2	none
Tennis 0.5B	600	1	600	1 $\times$ 1	none
Continued 7B	600	2	74	4 $\times$ 4	4-bit NF4

Optimisation Paged 8-bit AdamW; learning rate 2×10^{-4} ; cosine schedule; warmup ratio 0.03; weight decay 0; sequence limit 2,048; seed 42.
LoRA Rank 16; $\alpha = 32$; dropout 0.05; targets q , k , v , and o projections; no bias.
Checkpointing Enabled for TISER 7B and continued 7B; disabled for tennis 0.5B. The 4-bit run uses double quantisation.

rates, accumulation 4/8, and dropout 0/0.05), then fixed for one holdout evaluation. Retention compares the original (C0) and adapted (C1) adapters on 100 seed-42 items per original split. A 10,000-resample paired macro-EM interval triggers replay only if its upper bound is below -0.02 , stops as non-inferior if its lower bound exceeds -0.02 , and otherwise stops as inconclusive.

E. Conflict Sampling and Reflection Audit

Five recall samples use temperature 0.7/top- p 0.9; three-vote agreement with gold yields 571 eligible sources. Edits produce 203 C1, 520 C2, 333 C3, and 120 controls. The matrix uses one greedy decode per row. Seed-42 row bootstraps (2,000 resamples) and paired McNemar tests do not cluster variants from the same source, so uncertainty may be anti-conservative. A blinded two-pass GPT-5.6 Sol audit labels 2,295 unique nonempty reflections as explicit conflict, no conflict, or unscorable; exact evidence is required for positives and all 66 disagreements are adjudicated. An earlier unverifiable Claude aggregate is excluded.

F. Compute and Software

Training used one H200 (bf16, about 6.5 h); vLLM [7] evaluation used one H100 (22,014 items in about 17 min). Versions are PyTorch 2.4.1/CUDA 12.4, Transformers 4.46.3, PEFT 0.13.2, TRL 0.12.2, and vLLM 0.22.0/PyTorch 2.11/CUDA 13.0.

V. Results

A. Baseline Reproduction

Macro-EM is 0.033 below the paper and macro-F1 is 0.005 higher (Table II), indicating similar aggregate performance but a different TGQA error profile.

TGQA’s 0.571 EM versus 0.870 F1 suggests partial or formatting errors, but the 0.274 paper gap remains unexplained; unavailable row annotations preclude a defensible adjusted score.

TABLE II
In-domain reproduction; ToT-semantic (OOD EM 0.381) is excluded.

Split	n	Ours		Paper	
		EM	F1	EM	F1
TGQA	3,316	0.571	0.870	0.845	0.942
TempReason L2	5,397	0.907	0.929	0.855	0.875
TempReason L3	4,426	0.961	0.974	0.915	0.949
TimeQA easy	2,997	0.980	0.989	0.979	0.983
TimeQA hard	3,078	0.970	0.981	0.961	0.972
Macro (5)	19,214	0.878	0.949	0.911	0.944

TABLE III
TISER-selected conflicts ($n = 1,056$) and separate controls ($n = 120$). Orig.-EM scores the pre-edit answer; Δ is relative to base+standard.

Condition	Faith-EM	Faith-F1	Orig.-EM	Δ EM	Ctrl. EM
Base + standard	0.348	0.579	0.295	—	0.658
Base + TISER	0.525	0.718	0.228	+0.176	0.883
TISER + standard	0.560	0.759	0.251	+0.211	0.700
TISER + TISER	0.774	0.914	0.153	+0.425	0.900

B. Context-Memory Conflict

On 1,056 conflicts, TISER weights+prompt achieve 0.774 faithful EM versus 0.348 for base+standard (Table III). Same-model prompting adds 0.214; same-prompt TISER weights add 0.249, although TISER-based eligibility confounds the latter.

Including controls, EM is 0.380/0.561/0.574/0.787 in table order. Across all 1,176 rows, prompt/model-axis gains are 0.213/0.225 ($p < 10^{-40}$), but row dependence makes effect sizes, not these p -values, primary.

C. Conflict Type and Reflection

C1/C2 are largely solved, whereas C3 is the only class where memorised EM exceeds faithful EM (Table IV); transformation differences and conspicuous C2 distractors preclude a causal diagnosis.

Explicit acknowledgement occurs in 87/1,053 base+TISER and 36/1,056 TISER+TISER conflicts, versus 3/120 and 3/119 controls. For TISER+TISER, 29/36 flags occur in C3 and 810/817 faithful answers are silent (99.14%, 95% CI 98.24–99.58%). Thus reflection rarely identifies conflict and does not clearly discriminate conflicts from controls; “override” is behavioural, not mechanistic.

D. Tennis-Domain Adaptation

At 0.5B, prompting does not help but tennis training adds 0.085 EM. At 7B, continued adaptation adds 0.147 EM/0.151 F1 over the original adapter. Missing unadapted 7B results prevent attributing direct-transfer performance to TISER.

Context-faithful exact match

	Model	Prompt format	
		Standard	TISER
	Base model	0.380 [0.352, 0.407]	0.561 [0.531, 0.589]
	TISER SFT	0.574 [0.546, 0.603]	0.787 [0.761, 0.810]

Value [95% bootstrap CI]; $n = 1,176$ per cell

Fig. 2. Faithful EM and row-bootstrap 95% intervals across model–prompt conditions ($n = 1,176$, controls included).

TISER SFT + TISER prompt

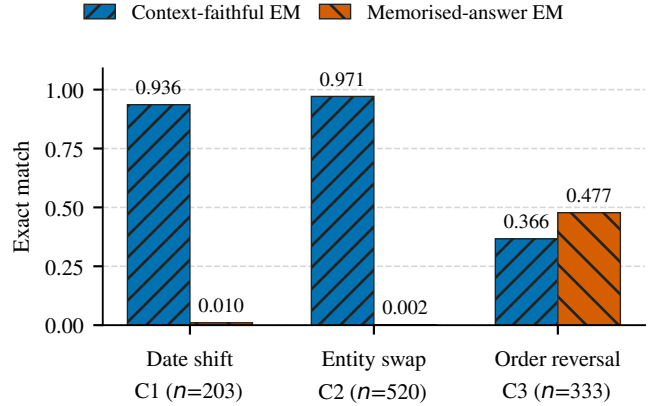


Fig. 3. Faithful versus memorised EM by conflict class for TISER weights+prompt.

E. Retention and Final Holdout

Retention changes by -0.004 macro-EM; its paired interval $[-0.028, 0.020]$ meets neither gate, so replay is not run. Separate OOD EM falls 0.410 to 0.320. On 113 supported holdout items, adaptation raises EM 0.575 to 0.735 ($\Delta = 0.159$, 95% CI 0.080–0.239; McNemar $p = 0.000277$) and F1 0.677 to 0.834. Exploratory gains are largest for duration; small category samples are not confirmatory.

F. Tennis Audit Results

All 337 evaluation items are model-judge supported. The 18 semantic concerns occur in training (13 under-determined, four wrong-gold proposals, one inconsistent); 14 enter the reported 600 traces, but training files remain unchanged.

TABLE IV

Conflict results for TISER weights+prompt; Refl. is judge-flag rate.

Class	n	Faith-EM	Mem-EM	Refl.
C1 date shift	203	0.936	0.010	0.020
C2 entity swap	520	0.971	0.002	0.006
C3 order reversal	333	0.366	0.477	0.087
All conflicts	1,056	0.774	0.153	0.034

TABLE V

Reflection audit: explicit conflict flags, A/B disagreement, and faithful answers without a flag (Silent).

Condition	Conflict	Control	Disagree	Silent
Base + TISER	87/1,053	3/120	41/1,173	543/554
TISER + TISER	36/1,056	3/119	25/1,175	810/817

Among the 600 training traces, 582 are supported, 14 have unsupported reasoning, and four contradict the answer. Overall, 579/600 pass both audits and 21 have at least one concern. Without retraining, their causal effect is unknown; model judgments are not human certification.

VI. Discussion and Limitations

The reproduction matches the aggregate regime, not the full error profile or original ablations. Conflict results are descriptive: TISER-based selection weakens vanilla-model comparisons; C2 can be conspicuous; and row-level statistics ignore shared source items. EM is format-sensitive, greedy decoding gives one response per condition, and single-family model audits lack human calibration and reproducible backend settings. Tennis evidence is limited to synthetic data, one model family, one training run per adapter, and 600 traces containing 21 audited concerns. The missing unadapted 7B baseline blocks attribution to prior TISER training; the 224-item selection set and 113-item holdout limit precision, and holdout history cannot be externally verified. Retention is inconclusive, so replay effectiveness is untested.

VII. Conclusion

We reproduce TISER’s aggregate regime but retain an unexplained TGQA gap. On a TISER-selected benchmark, TISER weights and prompting increase contextual adherence, while explicit reflection remains rare and order reversal weak. Continued tennis adaptation improves holdout EM from 0.575 to 0.735 over the original adapter, but the absent unadapted 7B baseline limits attribution. Retention is inconclusive. Context use, conflict acknowledgement, and adaptation should therefore be evaluated separately.

TABLE VI

Tennis selection results ($n = 224$); deltas use each size block’s first row.

Condition	EM	F1	Δ EM	Δ F1	Malf.
Qwen2.5-0.5B-Instruct					
Base, standard prompt	0.379	0.472	—	—	0
Base, TISER prompt	0.375	0.420	−0.004	−0.052	5
Tennis-only adapter	0.464	0.516	+0.085	+0.045	0
Qwen2.5-7B-Instruct					
Original TISER	0.580	0.701	—	—	0
Continued adaptation	0.728	0.852	+0.147	+0.151	0

TABLE VII

Retention and holdout results; C1–C0, paired bootstrap 95% CIs.

Evaluation	C0	C1	Δ	95% CI
Retention macro EM	0.880	0.876	−0.004	[−0.028, 0.020]
Retention macro F1	0.946	0.936	−0.010	[−0.028, 0.007]
Tennis holdout EM	0.575	0.735	+0.159	[0.080, 0.239]
Tennis holdout F1	0.677	0.834	+0.157	[0.082, 0.232]

VIII. Reproducibility

The repository provides data, training, evaluation, audit, and statistical workflows with seed 42, configurations, predictions, metrics, prompts, hashes, provenance, and environment metadata. It retains row-level primary/adjudicated judgments and a resumable Colab retention/holdout workflow; hashes link each run to data, adapter, configuration, and scoring view.

TABLE VIII

Tennis audits: U underdetermined, WG wrong gold, I inconsistent,
UR unsupported reasoning, C contradictory.

Audit	n	Supported	Other final labels	A/B dis.
Semantic (unique)	1,121	1,103	13 U; 4 WG; 1 I	9
Trace	650	632	14 UR; 4 C	12

Appendix A

Tennis Data-Generation Prompts

The exact, verbatim prompts are versioned in the repository as `RAW_GENERATION_PROMPT.md` and `TRACE_GENERATION_PROMPT.md`. The compact excerpts below preserve the task, output contract, and principal constraints; the linked files are authoritative for exact reproduction.

A. Stage 1: Raw Example Creation

ABBREVIATED PROTOCOL (not the verbatim prompt)

Task: Generate batches of exactly 50 synthetic professional-tennis temporal-QA examples.

Output: one valid JSON array with records
{ "context": "...", "question": "...", "answer": "..." }

Requirements:

- Each self-contained context has 3--6 sentences and at least three temporally related events.
- The answer must be short, unique, and inferable only from context.
- Cover before/after, first/last, duration, overlap, and tournament progression questions, with varied difficulty and answer polarity.
- Avoid ambiguity, external knowledge, and near duplicates.
- Do not output reasoning, timelines, reflections, Markdown, or text outside the JSON array.

ChatGPT-5.5 used the full linked prompt to generate the 1,122 raw records before schema auditing, categorisation, and splitting. The omitted prompt text enumerates tennis scenarios, target category proportions, examples, and a pre-output quality checklist.

B. Stage 2: TISER Trace Generation

ABBREVIATED PROTOCOL (not the verbatim prompt)

Input: question_id, category, context, question, and gold_answer.

Output: one JSONL record per example:
{ "question_id": "...", "output": "..." }

Required output structure:

```
<reasoning>...  
<timeline>...</timeline>  
<reflection>...</reflection>  
...</reasoning>  
<answer>GOLD_ANSWER</answer>
```

Use only the supplied context; reason about the required temporal relations; construct a chronological timeline; verify the gold answer in the reflection; invent no event, date, entity, or relation; preserve question_id; and return JSONL only.

The full linked prompt generated the supervised traces. The final 600 records exactly match batches 1–12 in order. Nineteen malformed physical JSONL lines covering 26 records were repaired, and 200 outputs that lacked the requested XML-like tags were mechanically wrapped using their existing text and supplied gold answer. All 600 final rows pass structural and gold-answer-equality checks; those checks do not establish semantic entailment.

Appendix B

Model-Judge Prompt and Adjudication

The reflection, semantic, and trace audits used one shared task prompt, pasted unchanged into the separate Judge A, Judge B, and adjudicator tasks. Its exact text is versioned as `GPT56_SOL_AUDIT_TASK_PROMPT.md`; each task additionally received only its assigned JSON bundle containing the audit-specific, pass-specific rubric and blinded items.

ABBREVIATED PROTOCOL (not the verbatim prompt)

Use only the assigned bundle. Do not inspect the repository, another judge pass, previous decisions, or experiment outputs. Judge every item individually under its batch rubric without keyword or default-label heuristics.

For every item, return its unchanged audit_id and source_sha256, one allowed label, an item-specific rationale, and any required exact source quotation. Provide a corrected answer only for semantic wrong_gold. For reflection labels, distinguish explicit conflict from routine timeline checking and record the permitted mention kind.

Validate every response file against its batch identity, schema, item coverage, allowed labels, and exact-evidence requirements.

The excerpt states the audit contract; the linked file preserves the exact response schema, allowed mention kinds, interruption behaviour, and operator instructions used in all passes.

A. Adjudication

Judge A and Judge B were compared by item identity. Follow-up bundles contained the original source and both decisions for every disagreement, plus every proposed wrong gold decision in the semantic audit. A third independent task, configured in the UI as gpt-5.6-sol with high reasoning effort, reapplied the relevant rubric without majority voting. The workflow adjudicated 66 reflection, 12 semantic, and 12 trace decisions. All 90 passed the label, identity, hash, and exact-quotation checks.

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